

# A Spatial-Domain Factor for Sparse-Sampling Circular-View Photoacoustic Tomography

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**Abstract**—Circular full-view configuration of photoacoustic imaging systems (C-PAI) has many applications in biomedicine (e.g., breast and brain imaging). To obtain a high-quality reconstructed image, dense spatial sampling (a large number of acoustic detectors) is needed, which makes the system expensive and challenging. Unfortunately, by reducing the number of spatial samples, streak artifacts appear, which degrade the quality of the reconstructed image. In this article, we propose a spatial-domain factor to suppress the streak artifacts and enhance the reconstructed image quality in a sparse sampling C-PAI system. Numerical and experimental studies are conducted to evaluate the proposed method. The results show that by reducing the number of spatial samples by one-fifth of the minimum required value to meet the Nyquist criteria, the proposed method provides a higher quality reconstructed image in terms of artifacts suppression and resolution improvement compared to the conventional method with dense spatial sampling. The proposed method improves the structural similarity index measure (SSIM), generalized contrast-to-noise ratio (gCNR), CNR, and tangential resolution values up to 100%, 9%, 38.6 dB, and ~45%, respectively. Based on the advantages of the proposed method, a low-cost version of a C-PAI system for clinical applications can be developed.

**Index Terms**—Circular full-view, computed tomography, photoacoustic imaging system, sparse sampling, streak artifact, tangential resolution.

## I. INTRODUCTION

**P**HOTOACOUSTIC (PA) imaging (PAI) is a nascent biomedical imaging modality with many clinical and pre-clinical applications [1], [2], [3], [4]. In a PAI system, a laser source irradiates the biological tissues. Then, based on the optical properties of the tissue, thermal expansion occurs. Subsequently, a pressure rise is induced in the medium which in turn

propagates as a PA wave in all directions [5], [6]. The generated PA waves are captured by ultrasound detectors. The more solid angles the detectors cover during data acquisition, the better is the resolution and contrast of the reconstructed image [7], [8]. Therefore, among different detector configurations of a PAI system, the resolution and contrast of the full-view cylindrical/circular configuration of a PAI system (C-PAI) is better than others [8], [9].

To obtain a high resolution/contrast image in a circular configuration, small aperture detectors with dense-spatial sampling (hundreds to thousands number of measurements) are used in the imaging system [10], [11]. However, (1) small aperture transducers are expensive, (2) manufacturing the ultrasound array with hundreds of densely arranged transducers would be expensive and challenging, and (3) if a single element ultrasound transducer (SUT) is used to acquire data, hundreds to thousands number of measurements should be obtained by simply rotating the SUT around the sample a full 360 degree, resulting in longer data acquisition time. On top, if the SUT has large aperture, due to their limited field of view (FOV), the resolution and contrast degrade as the off-center distance increases [12], [13]. Moreover, streak artifacts appear in a sparse-sampling PAI system [14], [15]. To compensate for the effect of large aperture transducer and suppress streak artifacts in a sparse-sampling PAI system several algorithms and methods have been reported.

To compensate for the large aperture effect, ring-shaped arrays are developed for a C-PAI system [16], [17]. Kalva and Pramanik [18] proposed a modified delay-and-sum beamformer to improve tangential resolution at the off-center distance. However, by using their method, the reconstructed image will become blurred. Some advanced back-projection-based methods are presented to improve the tangential resolution in the off-center distances [19]. The computational cost of this method becomes high in a large region of interest (ROI). Recently, a multi-angle data acquisition structure is developed to achieve a location-independent resolution in the imaging region [20], [21]. Also, the virtual point detectors concept with focused ultrasound transducers is proposed to compensate for the effect of a large aperture detector [22], [23]. Since the sensitivity of focused transducers is reduced in out-of-focus regions, more streak artifacts appear in the images [24]. Rajendran and Pramanik [25] also proposed a deep learning approach to improve the tangential resolution. Such methods need large number of data sets for training the network and can be computationally heavy.

Manuscript received 23 September 2022; accepted 12 December 2022. Date of publication 19 December 2022; date of current version 29 December 2022. This work was supported by Iran Natural Sciences Foundation under Grant 97018266. (Corresponding author: Zahra Kavehvasht.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSTQE.2022.3229622>.

Digital Object Identifier 10.1109/JSTQE.2022.3229622

Several studies have been reported to remove aliasing and streak artifacts. Hu et al. [10] proposed a spatiotemporal antialiasing algorithm to mitigate streak artifacts in PAT. In their approach, radius-dependent low pass filter and spatial interpolation are applied to remove aliasing and suppress artifacts. Although this method mitigates streak artifacts, it causes blurring. Cai et al. [26] proposed an adaptive back-projection algorithm for streak artifacts suppression. The parameters of their reconstruction method depend on the properties of the tissue. Therefore, achieving a high-resolution image in some cases may be challenging. Coherence-based weighting factors are presented to suppress artifacts in C-PAI systems [27], [28]. Model-based reconstruction algorithms are also presented to improve image quality and eliminate artifacts and background noise [29], [30]. Some deconvolution algorithms are presented to enhance the reconstructed image quality. Awasthi et al. [31] developed an image-guided filter to improve the reconstructed image. Van de Sompel et al. [32] compared three temporal data deconvolution filters (Fourier, Weiner, and Tikhonov). Since the matrix model is considered and is stored in the memory in these methods, for a large ROI with a small grid size, the size of this matrix becomes too large and possibly leads to high computational costs. Some other iterative image restoration methods total-variation and LSQR are also proposed to improve reconstructed image quality. However, iterative algorithms are generally time-consuming and computationally expensive [33]. Deep learning (DL) methods are also presented extensively to improve contrast and image quality [34]. To eliminate noise and improve the signal-to-noise ratio (SNR) Awasthi et al. [35] proposed a deep neural network. Lu et al. [36] presented an artifact removal technique with an unsupervised method. Davoudi et al. [37] proposed a DL method to eliminate streak artifacts for sparse data. Although DL algorithms improve the quality of the reconstructed image, the lack of large experimental datasets limits their performance when applied on *in vivo* scenarios. Compressed sensing (CS) is a new iterative technique, which reconstructs a high-quality image from sparse data. However, the image reconstruction process of CS-based methods is slower than the conventional back projection methods [14]. Recently, Gao et al. [38] accelerated the image reconstruction speed by parallelization on a GPU.

In this work, we propose a conceptually different method with low computational cost to solve both the mentioned issues. Our method suppresses streak artifacts and improves the resolution/contrast at the off-center distance without blurring. In the proposed method, a standard deviation-based (STD) method is developed to separate the map of artifacts and PA absorbers in the reconstructed image. First, an initial image is reconstructed by performing a back-projection algorithm on all captured PA signals by the detectors [7], [39], [40]. Then, for all pixels in the ROI of the initial reconstructed image, the STD values of all back-projected signals (spatial-domain signals) in all pixels are calculated. Since the amplitude of back-projected signals of two detectors facing each other are in the opposite sign in all pixels, the higher the value of the STD in a pixel, the higher possibility that the pixel refers to an excited PA absorber. For more accuracy in separating the artifacts and PA absorbers map,

a weighting factor is applied to the calculated STD map. For each pixel, this weighting is calculated based on the STD values of that pixel's neighborhood in the initial reconstructed image. Finally, a high-quality reconstructed image with a low number of spatial samples is obtained by applying the spatial-domain (proposed) factor to the reconstructed image. The results show that the proposed factor separates, perfectly the artifact and PA absorber map in the reconstructed image, which results in suppressing artifacts and improving the image quality.

The rest of this paper is organized as follows. Materials and methods are explained in Section II. Numerical and experimental results are provided in Sections III. Discussions and Conclusion are presented in Section IV and V.

## II. MATERIALS AND METHODS

### A. Image Formation in a C-PAI System

As a laser source irradiates a PA absorber at location  $\vec{r}$ , due to the photoacoustic effect, a pressure rise ( $p_0(\vec{r})$ ) is induced and propagates in the medium in all directions. In thermal confinement, the acoustic pressure at location  $\vec{r}$  and time  $t$  is as follows [41]:

$$p(\vec{r}, t) = \frac{1}{4\pi v_s^2} \frac{\partial}{\partial t} \left[ \frac{1}{v_s t} \int d\vec{r}' p_0(\vec{r}') \delta \left( t - \frac{|\vec{r} - \vec{r}'|}{v_s} \right) \right], \quad (1)$$

where,  $p(\vec{r}, t)$  is the acoustic pressure at location  $\vec{r}$  and time  $t$  and  $v_s$  is the speed of sound. By considering  $N$  acoustic detectors in a C-PAI system, the acoustic pressure captured by each detector ( $n$ ) would be:  $p_n(t) = p(\vec{r}_n, t)$ , where,  $\vec{r}_n$  determines the vector that connects the location of the PA absorber to the location of  $n^{\text{th}}$  detector that located at  $(r_n, \theta_n)$  [or  $(x_n, y_n)$  in Cartesian coordination]. The basic principle of a C-PAI system is depicted schematically in Fig. 1 (a). To reconstruct the initial pressure at location  $(r', \theta')$  [or  $(x', y')$  in Cartesian coordination], the relevant delayed pressure captured by all detectors have been generated on an arc-shaped surfaces [see gold and green signals and the intersection of the arcs in Fig. 1(a)] then converged at that location  $(r', \theta')$ . This procedure is the basis of the back-projection (BP) algorithm. The output of the BP algorithm for location  $(r', \theta')$  in Polar coordination and  $(x', y')$  in Cartesian coordination is given as follows:

$$p_{BP}(r', \theta') = p_{BP}(x', y') = \sum_{n=1}^N p_n(t_n), \quad (2)$$

where,  $t_n = \frac{|\vec{r}_n - \vec{r}'|}{v_s} = \frac{\sqrt{(x_n - x')^2 + (y_n - y')^2}}{v_s}$  is time of flight (TOF) from the initial pressure to the  $n^{\text{th}}$  detector.

### B. Proposed Method

1) *Pressure Distribution Propagation of an Excited PA Absorber*: If a sphere with a radius of  $R_s$  and centered at  $r = 0$ , is irradiated by a nanosecond laser pulse, the distribution of the initial pressure rise  $p_0$  is derived from (1). If the detector is located outside of the sphere ( $r > R_s$ ), the pressure distribution

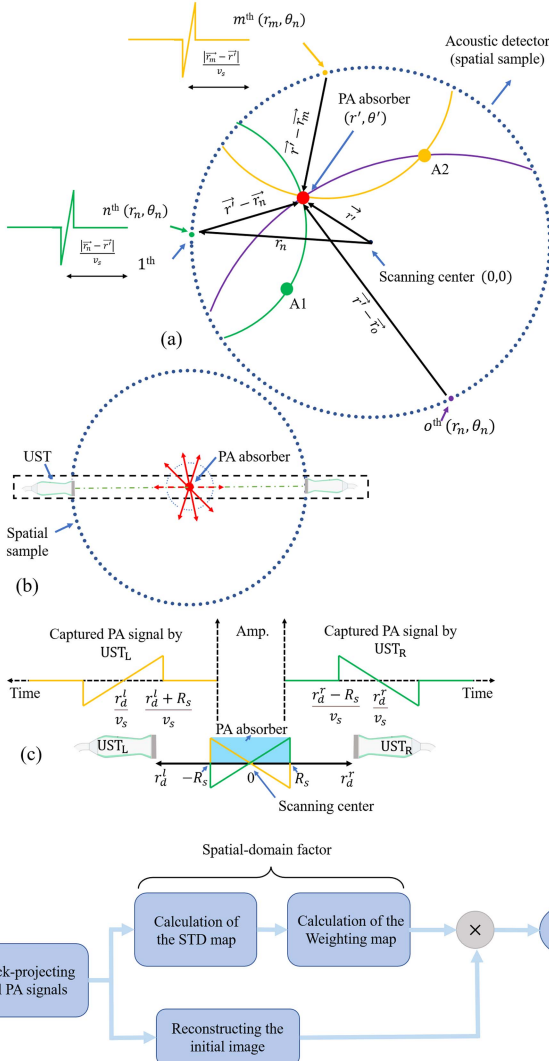


Fig. 1. (a) The schematic of principle of a C-PAI system.  $n$  acoustic detectors are placed around the imaging region. The red dot is an excited PA absorber. A1 and A2 indicate artifacts. Arc-shaped surfaces are the back-projected signals captured at time  $\frac{|\vec{r}_d - \vec{r}|}{v_s}$ .  $v_s$  is the speed of sound. (b) The schematic of a generated PA wave in a C-PAI system contains a pair of acoustic detectors. the red dot indicates the PA absorber. Double-headed arrows indicate solid angles of generated PA wave in several directions. Dashed double-headed arrow indicates the generated wave in the direction. Green dashed dot line indicates the horizontal direction. UST; ultrasound transducer. (c) Overlaying the one-dimensional plot of the back-projected signals of a pair of USTs facing each other and the ground truth (a light blue rectangle with a diameter of  $2 R_s$ ).  $r_d^r$  and  $r_d^l$  are the location of the right and left UST, respectively. Gold and green N-shaped signals are the captured PA signals by left (UST<sub>L</sub>) and right UST (UST<sub>R</sub>), respectively. (d) The workflow diagram of our method.

at distance  $r$  and time  $t$  would be as follows [41]:

$$p(r, t) = p_0 \left[ \begin{aligned} &U(R_s - v_s t - r) \\ &+ \frac{r - v_s t}{2r} U(R_s - |v_s t - r|) \\ &\times U(R_s + v_s t - r) \end{aligned} \right], \quad (3)$$

where,  $U$  is Heaviside function,  $r$  is the location of detector and  $t$  is TOF of PA wave between center of the sphere and detector. As the PA absorber is excited, the initial pressure

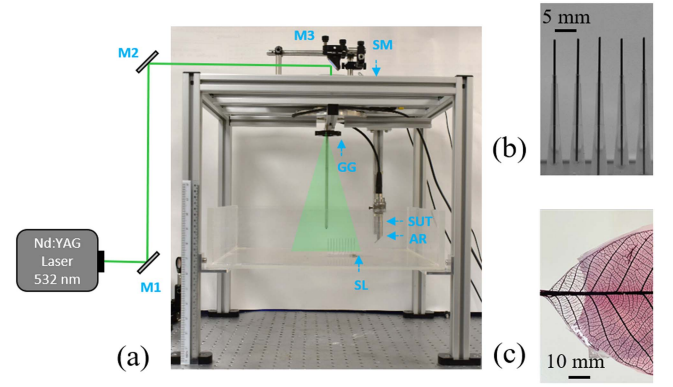


Fig. 2. (a) Photograph of the C-PAI system; SM, stepper motor; GG, ground glass diffuser; SUT, single-element ultrasound transducer; AR, Acoustic reflector; SL, sample; M1, M2, M3, mirror. (b) Five pencil leads with diameter of 0.5 mm. (c) Leaf skeleton phantom.

begins to propagate in all directions. The propagation of pressure wave in several directions is depicted in Fig. 1(b) as several double-headed arrows. Each double-headed arrow is interpreted as a solid angle pair of the generated PA waves that propagate along a specific dimension. Therefore, each pair of solid angles of PA waves are captured by two acoustic detectors facing each other [see the dashed rectangle in Fig. 1(b) for the solid angle pair of generated PA wave in the horizontal direction]. For simplicity, at first, to drive the theory, one pair of solid angles are assumed to propagate [e.g., see green dashed dot and dashed double-headed arrows in Fig. 1(b)]. Based on this assumption, as the initial pressure began to propagate, it divides into two equal parts, which propagate in opposite directions, left and right [see Fig. 1(b, c)]. If we consider the initial pressure of one part that begins to propagate to the right, as  $p_0(r) = p_0 U(r)U(-r + R_s)$  for  $0 \leq r < R_s$ , the bipolar pressure distribution propagating to the right,  $p(r, t)$ , will be as follows [41], [42]:

$$p(r, t) = p_0 \left[ \begin{aligned} &\frac{r + v_s t}{2r} p_0(r + v_s t) \\ &+ \frac{r - v_s t}{2r} p_0(-r + v_s t) \\ &+ \frac{r - v_s t}{2r} p_0(r - v_s t) \end{aligned} \right]. \quad (4)$$

In (6), the first term ( $\frac{r + v_s t}{2r} p_0(r + v_s t)$ ) represents a converging spherical wave and other terms represent diverging spherical waves. By substituting  $p_0(r) = p_0 U(r)U(-r + R_s)$  in (6) and  $r$  by  $r_d^r$ , the pressure recorded by the detector at location  $r_d^r$  (see right detector in Fig. 1(c)) and time  $t$  is given in (7).

$$p_{r_d^r}(t) = p_0 \left[ \begin{aligned} &\frac{r_d^r + v_s t}{2r_d^r} U(r_d^r + v_s t) U(-r_d^r - v_s t + R_s) \\ &+ \frac{r_d^r - v_s t}{2r_d^r} U(-r_d^r + v_s t) U(r_d^r - v_s t + R_s) \\ &+ \frac{r_d^r - v_s t}{2r_d^r} U(r_d^r - v_s t) U(-r_d^r + v_s t + R_s) \end{aligned} \right]. \quad (5)$$

To calculate the pressure distribution that propagates to the left, we should substitute  $p_0(r)$  by  $p_0 U(r + R_s)U(-r)$ ,  $t$  by  $-t$  and  $r$  by  $r_d^l$  in (6). Therefore, the bipolar pressure recorded by the detector at location  $r_d^l$  and time  $t$  (see left detector in Fig. 1(c))



will be obtained as follows:

$$p_{r_d^l}(t) = p_0 \left[ \begin{aligned} & \frac{r_d^l - v_s t}{2r_d^l} U(r_d^l - v_s t + R_s) U(-r_d^l + v_s t) \\ & + \frac{r_d^l + v_s t}{2r_d^l} U(-r_d^l - v_s t + R_s) U(r_d^l + v_s t) \\ & + \frac{r_d^l + v_s t}{2r_d^l} U(r_d^l + v_s t + R_s) U(-r_d^l - v_s t) \end{aligned} \right] \quad (6)$$

By considering only two detectors at locations  $r_d^l$  and  $r_d^r$  in a C-PAI system, the initial pressure of a point at location  $r$ , based on BP algorithm would be as follows:

$$p_{BP}(r) = p_{r_d^l}(t_{r_d^l, r}) + p_{r_d^r}(t_{r_d^r, r}), \quad (7)$$

where,  $t_{r_d^l, r} = \frac{r - r_d^l}{v_s}$  and  $t_{r_d^r, r} = \frac{r - r_d^r}{v_s}$ . A plot of bipolar (positive followed by negative) pressure distribution  $p(r, t)$  for both left and right detectors are depicted in Fig. 1(c). By substituting (7) and (8) in (9) and considering  $r$  to be located in the ROI ( $r_d^l \leq r < r_d^r$ ), the amplitude of the reconstructed PA pressure is as follows:

$$p_{BP}(r) = p_0 \left[ \begin{aligned} & \left( \frac{\frac{+r}{2r_d^r} U(-r) U(r + R_s)}{+ \frac{r}{2r_d^r} U(+r) U(-r + R_s)} \right) \\ & + \left( \frac{\frac{r}{2r_d^l} U(-r + R_s) U(+r)}{+ \frac{r}{2r_d^l} U(r + R_s) U(-r)} \right) \end{aligned} \right]. \quad (8)$$

Since  $r_d^l = -r_d^r$ , the amplitude of  $p_{r_d^l}$  and  $p_{r_d^r}$  in (9) are the same with opposite signs. Given that the mean value of these two signals is close to zero, their STD value will become the sum of their squares. Therefore, the STD value of back-projected signals in locations that refer to the PA absorber [e.g., see the blue rectangle in Fig. 1(c)] is more than in other locations. It should be however noted that, in practice, the value of  $p_{BP}(r)$  is not exactly equal to zero. This is due to the attenuation coefficient of the medium and non-ideal conditions (such as acoustic heterogeneity, spatial sampling mismatch). Moreover, the generated PA wave by some geometric shapes (e.g., cylinder) is not exactly N-shaped and is asymmetric. Therefore, the amplitude of the back-projected signals from different detectors does not cancel each other completely.

A novel spatial-domain factor is presented in the following subsection to improve reconstructed image quality in terms of resolution and contrast by taking advantage of the mentioned effects.

2) *Spatial-Domain Factor*: As stated in the previous subsection, the amplitude of back-projected signals at location ( $r$ ) of each pair of detectors facing each other are of opposite sign. The STD value of this pair of detectors at the location ( $r$ ) is calculated as follows:

$$STD_{1D, PB}(r) = p_0 \sqrt{\begin{aligned} & \left( \frac{\frac{+r}{2r_d^r} U(-r) U(r + R_s)}{+ \frac{r}{2r_d^r} U(+r) U(-r + R_s)} \right)^2 \\ & + \left( \frac{\frac{r}{2r_d^l} U(-r + R_s) U(+r)}{+ \frac{r}{2r_d^l} U(r + R_s) U(-r)} \right)^2 \end{aligned}}. \quad (9)$$

Since  $r_d^l = -r_d^r$ , (11) is summarized as follows:

$$STD_{1D, PB}(r) = p_0 \sqrt{2 \left( \frac{\frac{+r}{2r_d^r} U(-r) U(r + R_s)}{+ \frac{r}{2r_d^r} U(+r) U(-r + R_s)} \right)^2}. \quad (10)$$

(12) shows that the STD value of each pair of signals that refer to the location of an excited PA absorber ( $R_s \leq r \leq R_s$ ) is a positive value. But the STD values of locations referring to other parts (e.g., background) are insignificant [see gold and green signals in the blue rectangle in Fig. 1(c)]. Therefore, the map of the STD value separates the PA absorber map and artifacts in a 1-D ROI.

By generalizing the mentioned pair of detectors (1-D propagation of generated PA wave) to all detectors (2-D propagation of generated PA wave), the measured STD of all pairs ( $STD_{2D, PB}$ ) at location ( $x', y'$ ) would be as follows:

$$STD_{2D, PB}(x', y') = \sqrt{\frac{\sum_{n=1}^N (p_n(t_n) - \frac{1}{N} \sum_{n=1}^N p_n(t_n))^2}{N}}, \quad (11)$$

where,  $t_n = \frac{\sqrt{(x_n - x')^2 + (y_n - y')^2}}{v_s}$ . The higher the value of the  $STD_{2D, PB}$  in a pixel, the higher possibility that the pixel refers to an excited PA absorber. Therefore, by applying  $STD_{2D, PB}$  to the reconstructed image, it expected that the artifacts to be suppressed. However, due to the non-directional nature of the back-projection procedure in 2-D back-projection (In back-projection the signal captured at time  $t$  has been back-projected on an arc-shaped surface of radius  $v_s t$ ), which results in appearing streak artifacts, the value of STD may be high in some locations of the ROI that refer to artifacts [e.g., see A1 and A2 in Fig. 1(a)]. Still, the intensity of pixels referring to the PA absorber is higher than others in the output of the BP. Thus, the STD value of neighboring pixels referring to the PA absorber is higher than those referred to the artifacts. Hence, for more accuracy in separating artifacts and PA absorbers' maps, a weighting factor should be applied to the  $STD_{2D, PB}$  map at each pixel. This weighting factor which is given in (14) calculates the STD value of the BP output of each pixel's neighbors. The diameter of this neighborhood square is around the central wavelength of the C-PAI system's detectors ( $\lambda_c$ ) which is the approximate resolution of the system.

$$W(x', y') = \sqrt{\frac{\sum_{i=-a}^a \sum_{j=-a}^a (p_{BP}(x' + i, y' + j) - \frac{1}{N'} \sum_{i=-a}^a \sum_{j=-a}^a p_{BP}(x' + i, y' + j))^2}{N'}}, \quad (12)$$

where,  $N' = 2a \times 2a$  is the number of pixels within the neighborhood.

By multiplying (13) by (14), the proposed spatial-domain factor ( $p_{spatial}$ ) will be obtained for pixel ( $x', y'$ ), which is given in (15). If a pixel in the ROI refers to a normal or weak artifact, both the  $STD_{2D, PB}$  and  $W$  values are low in comparison to the pixels referring to an excited PA absorber. Also, if pixels refer to strong artifacts, the multiplication of the  $STD_{2D, PB}$  and  $W$  values are insignificant compared to PA absorber pixels. Therefore, the  $p_{spatial}$  suppresses artifacts and keeps the PA

amplitudes.

$$p_{spatial}(x', y') = \sqrt{\frac{\sum_{n=1}^N (p_n(t_n) - \frac{1}{N} \sum_{n=1}^N p_n(t_n))^2}{N}} \times \sqrt{\frac{\sum_{i=-a}^a \sum_{j=-a}^a (p_{BP}(x' + i, y' + j) - \frac{1}{N'} \sum_{i=-a}^a \sum_{j=-a}^a p_{BP}(x' + i, y' + j))^2}{N'}} \quad (13)$$

By applying  $p_{spatial}$  to the output of BP algorithm at location  $(x', y')$ , the final reconstructed pressure pattern ( $p_{out}(x', y')$ ) would be obtained which is given in (15). It is expected that the streak artifacts are diminished, and the image quality is improved. This, in turn, results in a reduction in the number of acoustic detectors (spatial samples) without degradation of image quality.

$$p_{final}(x', y') = p_{spatial}(x', y') \times p_{BP}(x', y'). \quad (14)$$

The workflow diagram of our proposed method is shown in Fig. 1(d).

### C. Numerical and Experimental Studies

1) *Numerical Study*: The k-Wave MATLAB toolbox [43] was used to generate raw PA data for the numerical study. A complex vasculature phantom with dynamic alteration in amplitude was synthesized to evaluate the performance of suppressing artifacts of all methods. The speed of sound on the medium was considered as 1500 m/s. The ROI radius was considered 2.5 cm in a square medium (500×500 grids) with a grid size of 0.1 mm in both directions. Acoustic detectors have a center frequency of 2.25 MHz and 67% bandwidth. The minimum number of acoustic detectors (spatial samples) to avoid aliasing is  $N = \frac{8\pi R_{ROI}}{\lambda_c}$ , where  $\lambda_c$  is the higher cut-off frequency, and  $R_{ROI}$  is the radius of the ROI [10]. Based on system properties, the minimum number of acoustic detectors was calculated at 1200. To assay the performance of our method in suppressing artifacts, some other simulations were conducted with 240 and 480 acoustic detectors (one-fifth and two-fifth of the Nyquist criterion). To have a fair computing performance, all methods were measured on a desktop computer with a 64-bit CPU (i9, Intel, USA), 64 GB RAM, and a GPU board (GeForce GTX 2060, NVIDIA, USA).

2) *Experimental Study*: Two types of phantoms were used for the experiments: 1) pencil leads phantom (0.5 mm) containing five pencil leads placed at a distance of 0, 5, 10, 15, and 25 mm from the scanning center [Fig. 4(b)] to evaluate the resolution. 2) complex leaf phantom to evaluate the performance of our method in artifact removal [Fig. 4(c)]. A schematic diagram of the experimental setup [Fig. 4(a)] and a photograph of phantoms are shown in Fig. 4. A Q-switched Nd:YAG laser was used as an excitation source to deliver laser pulses of 10 Hz with a 5-ns laser pulse width at 532-nm wavelength. The scanning radius was considered 12 cm to ensure that all phantoms placed in the Fraunhofer zone of all transducers, and all transducers were placed far enough to capture all generated waves from each PA absorber. An unfocused ultrasound transducer with the center frequency of 2.25 MHz with ~67% nominal bandwidth

(Olympus NDT, V306-SU) and a 13 mm diameter active area fitted with an acoustic reflector (F102, Olympus NDT) was employed to acquire the PA waves around the ROI. The PA signals were amplified by a low noise amplifier (Olympus-NDT, 5072PR) and recorded with a 25 MHz sampling rate with a data acquisition (DAQ) card (GaGe, compuscope 4227) and stored in a desktop computer (IntelXeon, 3.7 GHz 64-bit processor, 16 GB RAM).

3) *Evaluation Metrics*: To assay, the performance of our method in suppressing artifacts in sparse data and also to compare with existing methods, some image quality metrics are calculated including the contrast-to-noise-ratio (CNR), structural similarity index measure (SSIM), and generalized contrast-to-noise ratio (gCNR) [44]. Moreover, the full-width-at-half-maximum (FWHM) of 1-D plots at different locations of ROI are also demonstrated.

4) *Compared With Existing Methods*: Since this paper proposes a spatial-domain factor to suppress artifacts and improve resolution, the combination of the proposed factor with DL/CS methods can further achieve better results. Therefore, for comparison with the existing methods, we did not apply the DL/CS methods. The results of numerical and experimental studies of the proposed method have been compared with the results of the following methods. (1) conventional back-projection [Universal Back-Projection (UBP)] algorithm [7]. (2) interpolation-based method. The selected interpolation method is a spatiotemporal antialiasing method to improve the image quality proposed by Hu et al. [10]. In this method, first, they used spatial interpolation to remove aliasing that appears in the reconstructed image (Whittaker–Shannon interpolation). By applying spatial interpolation, the number of detection points is numerically doubled. Then to remove the resultant spatial aliasing, they applied a radius-dependent low-pass filter (third-order low-pass Butterworth and a sinc filter) to the temporal signals. However, the low-pass filter blurs the reconstructed image. (3) contamination-tracing back-projection (CTBP) method is proposed to mitigate streak artifacts by Cai et al. [26]. In the CTBP method, a weighing factor ( $W^{CB}$ ) is applied to each back-projected signal. This factor is the suppression weight used for minimizing the influences of contamination sources. Some parameters in  $W^{CB}$  should be adopted (e.g., " $a''$ ", " $w''_{win}$ ", and a threshold value) based on the properties of the imaging target. For example, if streak artifacts are serious in the initial reconstructed image, " $a''$ " and " $w''_{win}$ " should be selected with high and low values, respectively. Also, the threshold value should be adopted to achieve best results. To suppress as much as streak artifacts as possible, the parameters " $a''$ " and " $w''_{win}$ " are usually set within 0.2-0.4 and 0-0.3, respectively. In the numerical and experimental studies, we set the parameters " $a''$ ", " $w''_{win}$ ", and the threshold value for the CTBP method, within 0.3, 0.15, and 0.5, respectively. These values are selected to achieve acceptable results.

## III. NUMERICAL AND EXPERIMENTAL RESULTS

### A. Numerical Results

1) *Vessel Phantom*: To assay the performance of the proposed method, different reconstruction methods have been tested

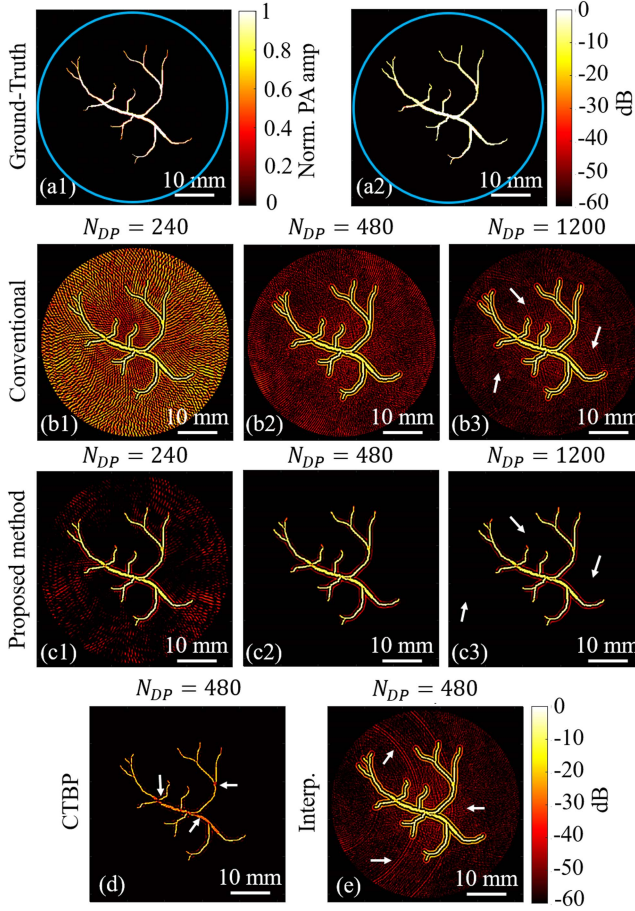


Fig. 3. The numerical results of the vessel phantom with dynamic alteration in amplitude. (a) and (b) show the vasculature phantom and the imaging region in amplitude and dB scale, respectively. (b1-b3) are the reconstructed image with the conventional back-projection method. (c1-c3) are the reconstructed image with the proposed method. (d) and (e) are the reconstructed image with the CTBP and interpolation method, respectively. In (b-e)  $N_{SS}$  is the number of acoustic detectors used in the simulation study.

on the vessel phantom. One simulation study with 1200 acoustic detectors (minimum number to meet Nyquist criteria) and two others with 480 and 240 acoustic detectors (spatial samples) were conducted. A schematic of the vessel phantom is shown in Fig. 3(a1, a2) in amplitude and dB scales. The results of 1200 acoustic detectors [see Fig. 3(b3, c3)] show that the phantom reconstructed well with both conventional (UBP) and the proposed methods. The SSIM values for the UBP and proposed method are 0.91 and 0.97, respectively. In the 1200 acoustic detectors scenario, some weak artifacts appear in the UBP results which are suppressed in the proposed method [see white arrow in Fig. 3(b3, c3)]. By reducing the number of acoustic detectors to two-fifth (480), more streak artifacts appear in the UBP result in comparison to 1200 acoustic detectors. Although the interpolation method suppresses streak artifacts in some locations, some strong artifacts remain [see white arrow in Fig. 3(e)]. The CTBP method suppresses all strong artifacts [see Fig. 3(e)]. However, some pixels that refer to strong PA absorbers are reconstructed with weak amplitude [see white arrows in Fig. 3(d)]. The proposed method suppresses streak artifacts

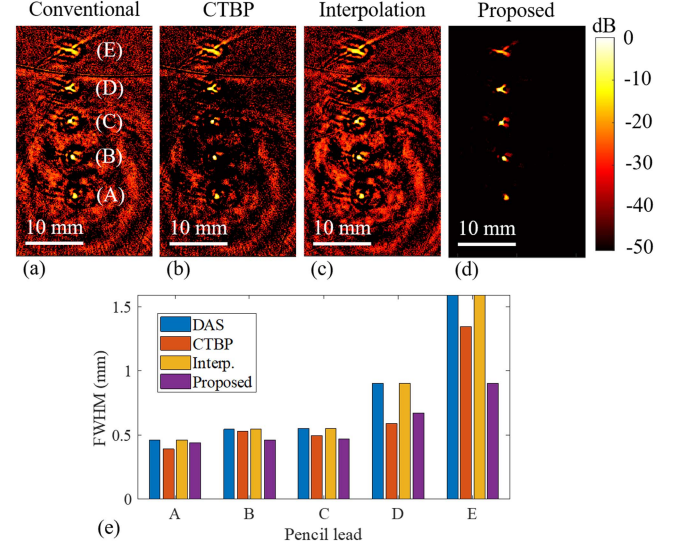


Fig. 4. The experimental results of the pencil lead targets. (a-d) are the reconstructed image with conventional, CTBP, interpolation, and the proposed method. The FWHM values of points (A-E) along the tangential direction in (a-d) are plotted (e).

without any missing data. The SSIM of the proposed method for the 480 acoustic detectors scenario was approximately equal to the 1200 detectors scenario (0.97). It means that reducing the number of acoustic detectors to two-fifths, would not degrade the reconstructed image in terms of SSIM value.

In 240 acoustic detectors cases (one-fifth of the minimum number of acoustic detectors to satisfy the Nyquist criteria), strong artifacts appear in the UBP result [see artifacts in Fig. 3(b1)]. Reducing the number of acoustic detectors to one-fifth results in degradation of the SSIM value from 0.91 to 0.48. In the 240 acoustic detectors case, the proposed method suppresses strong artifacts as much as in the 1200 acoustic detectors case. The SSIM degradation is insignificant and is almost constant. The SSIM and gCNR values of all mentioned methods with different acoustic detectors are given in Table I.

## B. Experimental Results

1) *Pencil Lead Targets*: Different reconstruction methods were tested on pencil lead targets. Five pencil leads separated by a distance of 5 mm, were placed from the scanning center to the edge of the ROI. The acoustic transducer employed to acquire PA waves is of a large aperture. All solid angles of each PA wave generated from excited PA targets located near the scanning center are captured by transducers (e.g., E, B, and C in Fig. 4). Therefore, the tangential resolution does not degrade and remains constant [see FWHM values of A, B, and C in Fig. 4(e)]. The FWHM for the PA absorber at location E (~2.5 cm from the scanning center) measured ~1.6 mm in UBP and interpolation methods. However, as the off-center distance increases, the captured solid angles of generated PA wave decrease. This results in degradation of tangential resolution and extension of the point spread function. The CTBP method compensates for the degradation of the tangential resolution in the off-center



TABLE I  
THE MEASURED QUANTITATIVE VALUES OF DIFFERENT TARGETS

| Figure of merit              | Vessel phantom (SSIM) |             |             | Vessel phantom (gCNR) |             |             | Leaf phantom (CNR) (dB) |             |             |
|------------------------------|-----------------------|-------------|-------------|-----------------------|-------------|-------------|-------------------------|-------------|-------------|
| Number of acoustic detectors | 240                   | 480         | 1200        | 240                   | 480         | 1200        | 240                     | 480         | 1200        |
| Conventional (UBP)           | 0.48                  | 0.89        | 0.91        | 0.78                  | 0.83        | 0.83        | 20.29                   | 22.44       | 23          |
| CTBP                         | 0.90                  | 0.95        | 0.95        | 0.60                  | 0.65        | 0.66        | 20.18                   | 22.6        | 23.15       |
| Interpolation                | 0.73                  | 0.91        | 0.92        | 0.71                  | 0.80        | 0.82        | 21.6                    | 22.8        | 23.4        |
| <b>Proposed method</b>       | <b>0.96</b>           | <b>0.97</b> | <b>0.97</b> | <b>0.85</b>           | <b>0.87</b> | <b>0.87</b> | <b>51.3</b>             | <b>54.4</b> | <b>61.6</b> |

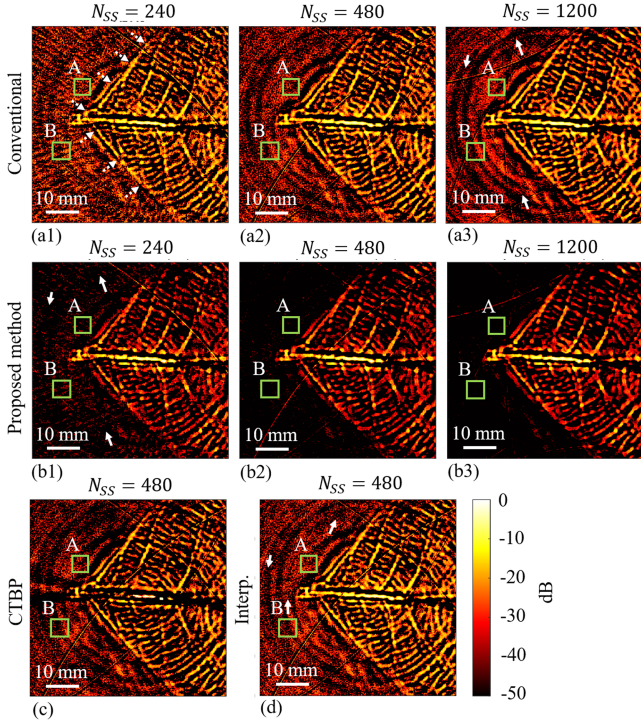


Fig. 5. The experimental results of the leaf phantom. (b1-b3) are the reconstructed image with the conventional back-projection method. (c1-c3) are the reconstructed image with the proposed method. (d) and (e) are the reconstructed image with the CTBP and interpolation method, respectively. In (b-e)  $N_{SS}$  is the number of spatial samples used in the experimental study.

distance and improves image quality. The FWHM of the reconstructed absorber at location E is measured as  $\sim 1.3$  mm, which is  $\sim 13\%$  better compared to UBP and interpolation. The CTBP method suppresses artifacts in the reconstructed image to some extent [see white arrows in Fig. 4(b)]. Still, in the proposed method, artifacts are suppressed more than CTBP and interpolation [compare arrows in Fig. 4(a-d)]. In the proposed method, the tangential resolution of the absorber at location E is measured as  $\sim 0.9$  mm, which is improved by  $\sim 45\%$  and  $\sim 33\%$  compared to UBP/interpolation and CTBP, respectively.

2) *Leaf Phantom*: A leaf phantom was used to analyze the performance of different reconstruction methods with 1200, 480, and 240 spatial samples (like numerical studies). The results of the reconstruction methods in the 1200 spatial samples case (Fig. 5) show that strong artifacts are completely suppressed by the proposed method in comparison to the UBP. The CNR

values in the leaf phantom was calculated as the ratio of the mean amplitude of the edge of the leaf phantom [see dashed arrows in Fig. 5(a)] to the mean of the STD values for two ROIs marked with green boxes in Fig. 5. The CNR values of 1200 spatial samples case are calculated at 23/23.15/23.4 and 61.6 dB for UBP/CTBP/interpolation and the proposed method, respectively. The proposed method improves the CNR by  $\sim 38.6$  dB. Although the CNR values of CTBP and interpolation are approximately the same, CTBP suppresses more artifacts than UBP in other locations [see white arrows in Fig. 5(c, d)]. The results of considering 480 spatial samples show that the proposed method suppresses artifacts better than other methods and the CNR values, reaches 22.44, 22.6, 22.8, and 54.4 dB for UBP, CTBP, interpolation, and the proposed method, respectively. Moreover, the image quality is maintained in the proposed method by reducing the number of spatial samples from 1200 to 240 [see Fig. 7(b)]. The CNR values for two ROIs marked with green boxes in Fig. 5 are calculated and shown in Table I.

#### IV. DISCUSSIONS

To obtain a reconstructed image to be free of artifacts and aliasing in a C-PAI system, dense spatial sampling should be done. Hence, if a ring-array transducer is used for acquiring PA data, it should have a small pitch and contains hundreds to thousands of transducers. Manufacturing a ring array with mentioned properties is challenging and expensive. Moreover, the data acquisition (DAQ) card with a large number of channels is also costly. If a single element transducer is used for data acquisition, capturing PA data with a large number of spatial samples, is time-consuming. Here, we introduced a novel spatial-domain factor, to enhance the performance of PA imaging. The main advantage of the proposed method is suppressing artifacts in a C-PAI system, especially in sparse-sampling cases. The minimum number of required spatial samples to meet the Nyquist criteria based on the system properties (central frequency, bandwidth, and radius of ROI) was calculated to be 1200. In general, the interpolation and CTBP methods improve the SSIM and CNR values and degrade gCNR values. The proposed method improves the SSIM and gCNR values more than other methods.

In the 1200 spatial samples scenario, the SSIM and gCNR values were measured as 0.91 and 0.83, respectively for the conventional (UBP) reconstruction algorithm. The image improvement by using the interpolation method in this scenario is insignificant. The CTBP method improves the SSIM value by  $\sim 5\%$  and degrades the gCNR value down to 0.66. The

SSIM and gCNR values reach 0.97 and 0.87, respectively. The improvement of SSIM and gCNR values in the proposed method are  $\sim 7\%$  and  $\sim 4\%$ , respectively. Moreover, the CNR values in the proposed method reaches 61.6 dB which is of  $\sim 38.6$  dB improvement compared to all previous methods (UBP, CTBP, and interpolation).

By reducing the number of spatial samples down to 480, the degradation in measured quantitative values is insignificant for all different methods. But by reducing the number of spatial samples down to one-fifth (240), the SSIM/gCNR values were degraded to 0.48/0.78 for the UBP reconstruction algorithm which is  $\sim 38\%/6\%$  compared to the 1200 spatial samples scenario. The SSIM/gCNR values of CTBP and interpolation methods in one-fifth spatial samples case reach 0.90/0.60 and 0.73/0.71, respectively. Although both the CTBP and interpolation methods improve the SSIM values  $\sim 87\%$  and  $52\%$  compared to UBP in the 240 spatial samples scenario, respectively, they degrade the gCNR values  $\sim 23\%$  and  $9\%$  compared to UBP, respectively. The proposed method improves both the SSIM and gCNR values by  $100\%$  and  $9\%$ , respectively. Moreover, the proposed method improves the CNR values by  $\sim 31$  dB compared to all previous methods in 240 spatial sample scenarios.

By comparing the SSIM and gCNR values of CTBP and interpolation methods in 240 spatial samples case with those values from UBP method in 1200 spatial samples case, it can be concluded that there is no improvement in SSIM and gCNR values. But the quantitative results of the proposed method indicate that the SSIM/gCNR values of the proposed method in 240 spatial sample cases are higher than the SSIM/gCNR values of the UBP method in 1200 spatial sample cases. This improvement could also be seen intuitively (compare arrows in Fig. 5(a3, b1)). Moreover, if a single-element transducer is used in the imaging system, the proposed method improves the tangential resolution by  $\sim 45\%$  at off-center distances.

The results show that a higher quality reconstructed image in terms of resolution and contrast with a lower number of spatial samples can be obtained by using the proposed method. Therefore, a low-cost version of C-PAI systems (e.g., brain and breast scanners) can be developed based on the proposed method's advantages. Since, the proposed method applies a factor to the initial reconstructed image, for future work, it can be combined with DL- and CS-based artifact removal methods for future improvements.

## V. CONCLUSION

In this work, we proposed a fast spatial-domain factor to suppress streak artifacts and improve resolution/contrast in a C-PAI system. The proposed method was evaluated using numerical and experimental studies. To assay the performance of the proposed method, two existing artifact removal methods were implemented, and the results are compared with our method. The results show that by reducing the number of spatial samples to one-fifth, the SSIM and gCNR values are almost constant and degradation is insignificant ( $\sim 2\%$ ) and are higher than other methods. The results showed that the quality of the reconstructed image with the proposed method with a sparse sampling

scenario (one-fifth) is better than the reconstructed image with the conventional method with dense spatial sampling. Moreover, due to the simplicity of implementing the proposed method to the conventional reconstruction method, the computational overhead of the proposed method is not significantly high and stands almost constant with the conventional back-projection method.

## REFERENCES

- [1] L. Li and L. V. Wang, "Recent advances in photoacoustic tomography," *BME Front.*, vol. 2021, pp. 1–17, 2021, doi: [10.34133/2021/9823268](https://doi.org/10.34133/2021/9823268).
- [2] J. Yao and L. V. Wang, "Perspective on fast-evolving photoacoustic tomography," *J. Biomed. Opt.*, vol. 26, no. 6, Jun. 2021, Art. no. 060602, doi: [10.1117/1.JBO.26.6.060602](https://doi.org/10.1117/1.JBO.26.6.060602).
- [3] D. Das, A. Sharma, P. Rajendran, and M. Pramanik, "Another decade of photoacoustic imaging," *Phys. Med. Biol.*, vol. 66, no. 5, Dec. 2020, Art. no. 05TR01, doi: [10.1088/1361-6560/abd669](https://doi.org/10.1088/1361-6560/abd669).
- [4] X. Hui, M. O. A. Malik, and M. Pramanik, "Looking deep inside tissue with photoacoustic molecular probes: A review," *J. Biomed. Opt.*, vol. 27, no. 07, 2022, Art. no. 070901, doi: [10.1117/1.JBO.27.7.070901](https://doi.org/10.1117/1.JBO.27.7.070901).
- [5] A. B. E. Attia et al., "A review of clinical photoacoustic imaging: Current and future trends," *Photoacoustics*, vol. 16, Dec. 2019, Art. no. 100144, doi: [10.1016/j.pacs.2019.100144](https://doi.org/10.1016/j.pacs.2019.100144).
- [6] S. Manohar and D. Razansky, "Photoacoustics: A historical review," *Adv. Opt. Photon.*, vol. 8, no. 4, pp. 586–617, 2016, doi: [10.1364/aop.8.000586](https://doi.org/10.1364/aop.8.000586).
- [7] M. Xu and L. V. Wang, "Universal back-projection algorithm for photoacoustic computed tomography," *Phys. Rev. E*, vol. 71, no. 1, 2005, Art. no. 016706, doi: [10.1103/physreve.71.016706](https://doi.org/10.1103/physreve.71.016706).
- [8] Y. Xu, L. V. Wang, G. Ambartsoumian, and P. Kuchment, "Reconstructions in limited-view thermoacoustic tomography," *Med. Phys.*, vol. 31, no. 4, pp. 724–733, Apr. 2004, doi: [10.1118/1.1644531](https://doi.org/10.1118/1.1644531).
- [9] S. Hakakzadeh and Z. Kavehvash, "Blind angle and angular range detection in planar and limited-view geometries for photoacoustic tomography," in *Proc. IEEE 29th Iranian Conf. Elect. Eng.*, 2021, pp. 922–926, doi: [10.1109/ICEE52715.2021.9544114](https://doi.org/10.1109/ICEE52715.2021.9544114).
- [10] P. Hu, L. Li, L. Lin, and L. V. Wang, "Spatiotemporal antialiasing in photoacoustic computed tomography," *IEEE Trans. Med. Imag.*, vol. 39, no. 11, pp. 3535–3547, Nov. 2020, doi: [10.1109/TMI.2020.2998509](https://doi.org/10.1109/TMI.2020.2998509).
- [11] Y. Xu, M. Xu, and L. V. Wang, "Exact frequency-domain reconstruction for thermoacoustic tomography. II. Cylindrical geometry," *IEEE Trans. Med. Imag.*, vol. 21, no. 7, pp. 829–833, Jul. 2002, doi: [10.1109/tmi.2002.801171](https://doi.org/10.1109/tmi.2002.801171).
- [12] M. Xu and L. V. Wang, "Pulsed-microwave-induced thermoacoustic tomography: Filtered backprojection in a circular measurement configuration," *Med. Phys.*, vol. 29, no. 8, pp. 1661–1669, 2002, doi: [10.1118/1.1493778](https://doi.org/10.1118/1.1493778).
- [13] M. Xu and L. V. Wang, "Analytic explanation of spatial resolution related to bandwidth and detector aperture size in thermoacoustic or photoacoustic reconstruction," *Phys. Rev. E*, vol. 67, no. 5, 2003, Art. no. 056605, doi: [10.1103/physreve.67.056605](https://doi.org/10.1103/physreve.67.056605).
- [14] X. Zhang et al., "Sparse-sampling photoacoustic computed tomography: Deep learning vs. compressed sensing," *Biomed. Signal Process. Control*, vol. 71, 2022, Art. no. 103233, doi: [10.1016/j.bspc.2021.103233](https://doi.org/10.1016/j.bspc.2021.103233).
- [15] S. Hakakzadeh, Z. Kavehvash, and M. Pramanik, "Artifact removal factor for Circular-view photoacoustic tomography," in *Proc. IEEE Int. Ultrason. Symp.*, 2022, pp. 1–4, doi: [10.1109/IUS54386.2022.9958228](https://doi.org/10.1109/IUS54386.2022.9958228).
- [16] L. Li et al., "Single-impulse panoramic photoacoustic computed tomography of small-animal whole-body dynamics at high spatiotemporal resolution," *Nature Biomed. Eng.*, vol. 1, no. 5, May 2017, Art. no. 0071, doi: [10.1038/s41551-017-0071](https://doi.org/10.1038/s41551-017-0071).
- [17] M. Nishiyama, T. Namita, K. Kondo, M. Yamakawa, and T. Shiina, "Ring-array photoacoustic tomography for imaging human finger vasculature," *J. Biomed. Opt.*, vol. 24, no. 9, 2019, Art. no. 096005.
- [18] S. K. Kalva and M. Pramanik, "Experimental validation of tangential resolution improvement in photoacoustic tomography using modified delay-and-sum reconstruction algorithm," *J. Biomed. Opt.*, vol. 21, no. 8, 2016, Art. no. 086011.
- [19] B. Wang, T. Ye, G. Wang, L. Guo, and J. Xiao, "Approximate back-projection method for improving lateral resolution in circular-scanning-based photoacoustic tomography," *Med. Phys.*, vol. 48 Apr. 2021, Art. no. 12, doi: [10.1002/mp.14880](https://doi.org/10.1002/mp.14880).



- [20] S. Hakakzadeh et al., "Finite transducer size compensation in two-dimensional photoacoustic computed tomography," in *Proc. IEEE Int. Ultrason. Symp.*, 2021, pp. 1–3, doi: [10.1109/IUS52206.2021.9593757](https://doi.org/10.1109/IUS52206.2021.9593757).
- [21] S. Hakakzadeh et al., "Multi-angle data acquisition to compensate transducer finite size in photoacoustic tomography," *Photoacoustics*, vol. 27, Sep. 2022, Art. no. 100373, doi: [10.1016/j.pacs.2022.100373](https://doi.org/10.1016/j.pacs.2022.100373).
- [22] L. Nie, Z. Guo, and L. V. Wang, "Photoacoustic tomography of monkey brain using virtual point ultrasonic transducers," *J. Biomed. Opt.*, vol. 16, no. 7, Jul. 2011, Art. no. 076005, doi: [10.1117/1.3595842](https://doi.org/10.1117/1.3595842).
- [23] M. Pramanik, G. Ku, and L. V. Wang, "Tangential resolution improvement in thermoacoustic and photoacoustic tomography using a negative acoustic lens," *J. Biomed. Opt.*, vol. 14, no. 2, Mar./Apr. 2009, Art. no. 024028, doi: [10.1117/1.3103778](https://doi.org/10.1117/1.3103778).
- [24] S. Hakakzadeh and Z. Kavehvasb, "Image quality equations for focused transducer in circular photoacoustic computed tomography," in *Proc. IEEE 29th Iranian Conf. Elect. Eng.*, 2021, pp. 917–921, doi: [10.1109/ICEE52715.2021.9544449](https://doi.org/10.1109/ICEE52715.2021.9544449).
- [25] P. Rajendran and M. Pramanik, "Deep learning approach to improve tangential resolution in photoacoustic tomography," *Biomed. Opt. Exp.*, vol. 11, no. 12, pp. 7311–7323, Dec. 2020, doi: [10.1364/BOE.410145](https://doi.org/10.1364/BOE.410145).
- [26] C. Cai et al., "Streak artifact suppression in photoacoustic computed tomography using adaptive back projection," *Biomed. Opt. Exp.*, vol. 10, no. 9, pp. 4803–4814, Sep. 2019, doi: [10.1364/BOE.10.004803](https://doi.org/10.1364/BOE.10.004803).
- [27] S. Hakakzadeh, S. M. Mostafavi, M. Amjadi, and Z. Kavehvasb, "Adapted coherent weighting in photoacoustic tomography," in *Proc. IEEE 28th Nat. 6th Int. Iranian Conf. Biomed. Eng.*, 2021, pp. 28–32.
- [28] S. Hakakzadeh and Z. Kavehvasb, "A Hilbert-based coherence factor for photoacoustic imaging," in *Proc. IEEE Int. Ultrason. Symp.*, 2022, pp. 1–3.
- [29] L. Ding, X. L. Dean-Ben, and D. Razansky, "Efficient 3-D model-based reconstruction scheme for arbitrary optoacoustic acquisition geometries," *IEEE Trans. Med. Imag.*, vol. 36, no. 9, pp. 1858–1867, Sep. 2017, doi: [10.1109/TMI.2017.2704019](https://doi.org/10.1109/TMI.2017.2704019).
- [30] M. A. Araque Caballero, J. Gateau, X.-L. Dean-Ben, and V. Ntziachristos, "Model-Based optoacoustic image reconstruction of large three-dimensional tomographic datasets acquired with an array of directional detectors," *IEEE Trans. Med. Imag.*, vol. 33, no. 2, pp. 433–443, Feb. 2014, doi: [10.1109/tmi.2013.2286546](https://doi.org/10.1109/tmi.2013.2286546).
- [31] N. Awasthi, S. K. Kalva, M. Pramanik, and P. Yalavarthy, "Image-guided filtering for improving photoacoustic tomographic image reconstruction," *J. Biomed. Opt.*, vol. 23, no. 9, 2018, Art. no. 091413.
- [32] D. Van de Sompel, L. S. Sasportas, J. V. Jokerst, and S. S. Gambhir, "Comparison of deconvolution filters for photoacoustic tomography," *PLoS One*, vol. 11, no. 3, 2016, Art. no. e0152597, doi: [10.1371/journal.pone.0152597](https://doi.org/10.1371/journal.pone.0152597).
- [33] C. Huang, K. Wang, L. Nie, L. V. Wang, and M. A. Anastasio, "Full-wave iterative image reconstruction in photoacoustic tomography with acoustically inhomogeneous media," *IEEE Trans. Med. Imag.*, vol. 32, no. 6, pp. 1097–1110, Jun. 2013, doi: [10.1109/tmi.2013.2254496](https://doi.org/10.1109/tmi.2013.2254496).
- [34] P. Rajendran, A. Sharma, and M. Pramanik, "Photoacoustic imaging aided with deep learning: A review," *Biomed. Eng. Lett.*, vol. 12, pp. 155–173, 2021, doi: [10.1007/s13534-021-00210-y](https://doi.org/10.1007/s13534-021-00210-y).
- [35] N. Awasthi, G. Jain, S. K. Kalva, M. Pramanik, and P. K. Yalavarthy, "Deep neural network-based sinogram super-resolution and bandwidth enhancement for limited-data photoacoustic tomography," *IEEE Trans. Ultrason., Ferroelect., Freq. Control*, vol. 67, no. 12, pp. 2660–2673, Dec. 2020, doi: [10.1109/TUFFC.2020.2977210](https://doi.org/10.1109/TUFFC.2020.2977210).
- [36] M. Lu et al., "Artifact removal in photoacoustic tomography with an unsupervised method," *Biomed. Opt. Exp.*, vol. 12, no. 10, pp. 6284–6299, Oct. 2021, doi: [10.1364/BOE.434172](https://doi.org/10.1364/BOE.434172).
- [37] N. Davoudi, X. L. Deán-Ben, and D. Razansky, "Deep learning optoacoustic tomography with sparse data," *Nature Mach. Intell.*, vol. 1, no. 10, pp. 453–460, 2019, doi: [10.1038/s42256-019-0095-3](https://doi.org/10.1038/s42256-019-0095-3).
- [38] M. Gao et al., "Graphics processing unit accelerating compressed sensing photoacoustic computed tomography with total variation," *Appl. Opt.*, vol. 59, no. 3, pp. 712–719, Jan. 2020, doi: [10.1364/AO.378466](https://doi.org/10.1364/AO.378466).
- [39] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvasb, "Fast adapted delay and sum reconstruction algorithm in circular photoacoustic tomography," in *Proc. IEEE 30th Int. Conf. Elect. Eng.*, 2022, pp. 49–54, doi: [10.1109/ICEE55646.2022.9827413](https://doi.org/10.1109/ICEE55646.2022.9827413).
- [40] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvasb, "Unipolar back-projection algorithm for photoacoustic tomography," in *Proc. IEEE Int. Ultrason. Symp.*, 2022, pp. 1–4, doi: [10.1109/IUS54386.2022.9957429](https://doi.org/10.1109/IUS54386.2022.9957429).
- [41] L. V. Wang and H. Wu, *Biomedical Optics: Principles and Imaging*. New York, NY, USA.: Wiley, 2012.
- [42] S. Hakakzadeh et al., "Signal restoration algorithm for photoacoustic imaging systems," *Biomed. Opt. Exp.*, vol. 2021, 2022, Art. no. 6689194.
- [43] N. A. Rejesh, H. Pullagurra, and M. Pramanik, "Deconvolution-based deblurring of reconstructed images in photoacoustic/thermoacoustic tomography," *J. Opt. Soc. Amer. Opt. Image Sci. Vis.*, vol. 30, no. 10, pp. 1994–2001, Oct. 2013, doi: [10.1364/JOSAA.30.001994](https://doi.org/10.1364/JOSAA.30.001994).
- [44] K. M. Kempinski, M. T. Graham, M. R. Gubbi, T. Palmer, and M. A. Lediju Bell, "Application of the generalized contrast-to-noise ratio to assess photoacoustic image quality," *Biomed. Opt. Exp.*, vol. 11, no. 7, pp. 3684–3698, Jul. 2020, doi: [10.1364/BOE.391026](https://doi.org/10.1364/BOE.391026).



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